What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Antibiotic Stewardship

Last updated June 13, 2025 9 min read CC BY-SA 4.0

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0:00 / 15:22 1x

Learning Objectives (2)

After completing this brick, you will be able to:

1. Describe the major targets, common mechanisms of how bacteria become resistant to drugs, and explain strategies to combat antimicrobial resistance.
2. Explain the rationale for the use of broad versus narrow spectrum and empiric versus targeted antibacterial therapy.

CASE CONNECTION

RP is a 72-year-old male who presents to the hospital with a fever, productive cough, and shortness of breath that has worsened over the past 48 hours. His medical history includes chronic obstructive pulmonary disease (COPD) and a recent hospital admission for a urinary tract infection, during which he received a 7-day course of ciprofloxacin.

On physical examination, RP is tachycardic, febrile (101.3°F), and has diminished breath sounds in the left lower lung field. A chest X-ray reveals consolidation in the left lower lobe, consistent with pneumonia. Based on his presentation, RP is started on empiric antibiotic therapy with a broad-spectrum beta-lactam and a macrolide. Blood and sputum cultures are sent to the microbiology lab. Two days later, the culture and sensitivity report indicates the growth of a multidrug-resistant (MDR) strain of *Klebsiella pneumoniae*, which is resistant to ciprofloxacin and the empiric beta-lactam but susceptible to carbapenems. RP is switched to a targeted regimen of meropenem.

RP is very confused as to why he has to switch antibiotics, and would like an explanation as to why he has to start a different antibiotic now? How would you explain resistance to RP? How would you explain empiric and definitive therapy to him? Keep these questions in mind as we go through this brick.

GO TO CONCLUSION ↓

From Acinetobacter to Toxoplasma species and Amoxicillin to Vancomycin, we have covered a TON of different bacteria and antibiotics. Knowing the most common pathogens for particular infections and their treatments will be important as you move forward. As you have learned, some bacteria are resistant, or may bec That is why the concept of antibiotic stewardship is so important. Antibiotic stewardship is a coordinated effort to improve the use of antibiotics to optimize patient outcomes while minimizing the development of antibiotic resistance and adverse drug events. It involves strategies aimed at ensuring that antibiotics are used appropriately. We will discuss some of those guiding principles here in this brick. As a soon to be practicing physician you have a duty to be a good steward over the antibiotics that you prescribe.

How do bacteria become resistant to antibiotics? (LO#1)

Perhaps the most significant reason to practice good antibiotic stewardship is to curb bacterial resistance to antibiotics. But why does limiting some antibiotic use, or only using certain antibiotics for certain infections help with resistance? It helps to take a look at some of the common bacterial mechanisms of resistance.

- *Intrinsic resistance*: The resistance is natural to the organism.
    - Example: *E. coli* is intrinsically resistant to vancomycin because this antibiotic is too large to penetrate the bacterial cell wall of *E. coli*.
- *Selection pressure*: Resistance occurs when antibiotics kill off susceptible bacteria, leaving behind more resistant strains to multiply.
    - Example: normal GI flora includes *Enterococcus*. When antibiotics, like oral vancomycin, eliminate susceptible *Enterococci*, vancomycin-resistant *enterococcus* (VRE) can become predominant.
- *Acquired resistance*: Bacterial DNA containing resistant genes can be transferred between different species and/ or picked up from dead bacterial fragments in the environment.
- *Enzyme inactivati* ak down the antibiotic. There are three primary to be aware of:
    - *Beta-lactamase* is produced by some bacteria and can break down the beta-lactam structure of beta-lactam antibiotics, rendering them ineffective. That is why many beta-lactam antibiotics are paired with beta-lactamase inhibitors, such as clavulanate, sulbactam, or tazobactam.
    - *Extended-spectrum beta-lactamases (ESBLs)* are beta-lactamases produced by some gram-negative bacteria that break down all penicillins and some cephalosporins. These can be difficult to kill and are usually treated with carbapenems.
    - *Carbapenem-resistant Enterobacteriaceae* (CRE) are multi-drug resistant gram-negative organisms that produce enzymes capable of breaking down penicillins, most cephalosporins, and carbapenems.

CLINICAL REASONING

Have you ever wondered why they are more multi-drug resistant organisms in the hospital? Or, why patients with hospital acquired infections, compared to community acquired, require broader-spectrum antibiotics?

Antibiotic use in the hospital setting, especially intravenous antibiotics, is much more prevalent. This allows bacteria in healthcare settings to gain resistance through selection pressure and acquired resistance based on consistent exposure to antibiotics.

In an effort to comba mented Antimicrobial Stewardship Programs (ASP). As mentioned above, ASPs are designed to optimize patient outcomes, minimize the development of resistance, and minimize adverse drug events.

ASPs consist of collaborative teams (e.g., Infectious Disease physicians and pharmacists, microbiology lab, and infection prevention personnel) that establish antibiotic guidance for their facility. They also conduct audits of prescribing habits and provide education to optimize practice and improve care. Some examples of ASP interventions include:

- Pharmacokinetic monitoring of some antibiotics (e.g., vancomycin and aminoglycosides)
- Use of clinical decision support software to rapidly identify pathogens and shorten time to starting effective treatment
- Restriction of select antibiotics (e.g., ceftaroline)
- Prospective audit and feedback to prescribers of selected antibiotics
- Timely transition from intravenous to oral antibiotics

CLINICAL CORRELATION

Ceftaroline is a good example of a restricted antibiotic. But why is it restricted? One reason is that it is the only fifth generation cephalosporin available, and it is unique because it has activity against a broad range of multi-drug resistant gram-positive organisms, including MRSA and VRE. So, in this instance it is restricted, or reserved, for certain infections with these known pathogens in order to limit exposure (i.e., bacterial resistance) and maintain coverage.

Mnemonic

Kill **E**ach **A**nd **E**very **S**trong **P**athogen

What is intrinsic resistance in bacteria?

Why are narrow preferred over broad spectrum? (LO#2)

In some of the bricks in this course we have covered the 'coverage' or 'spectrum of activity' for a variety of antibiotic classes. Some of those spectrums encompass a broad array of bacteria, while others cover fewer pathogens. This is the fundamental difference between narrow and broad-spectrum antibiotics. But, why would it be preferred to use narrow over broad spectrum? Since narrow-spectrum drugs target only specific types of bacteria, this minimizes disruption to the normal gut microbiome and *reduces the risk of developing antibiotic resistance* by only targeting the bacteria causing the infection. On the other hand, broad-spectrum antibiotics kill a wider range of bacteria, including beneficial ones, which can lead to increased resistance development when used unnecessarily.

Using narrow-spectrum antibiotics over broad-spectrum antibiotics is like using a precision screwdriver instead of a hammer to fix a delicate watch. The precision screwdriver targets the specific screws without damaging the intricate mechanisms inside, just as narrow-spectrum antibiotics target specific bacteria without disrupting the beneficial ones. On the other hand, using a hammer would cause unnecessary damage, similar to how broad-spectrum antibiotics can harm beneficial bacteria and increase resistance development when used unnecessarily. Therefore, just as a precision screwdriver is preferred for delicate tasks, narrow-spectrum antibiotics are preferred when the causative bacteria are known and can be specifically targeted.

How do empiric and targeted therapy differ?

Recall, from the 'Prin ssed empiric and definitive, or targeted, antibiotic therapies. *Empiric therapy* is our 'best guess' of antibiotic coverage for a patient with a proven or suspected infection, where an organism has not yet been identified. Of note, broad-spectrum antibiotics, as discussed above, can be warranted for empiric therapy, and they can be narrowed later in order to minimize pathogen exposure (i.e., decrease chance for resistance) and reduce adverse drug effects.

After culture and sensitivity results are known, the *targeted*, or definitive, treatment can begin. Unlike empiric therapy, with targeted therapy we know which organism(s) we are treating, and which antibiotics will work against them. At this point, it is important to be prudent, and a good antibiotic steward, to choose drugs that are safe, effective, narrow in spectrum, and cost effective. Figure 1 shows a a general approach to infectious diseases, and shows where empiric and definitive/ targeted therapies are initiated.

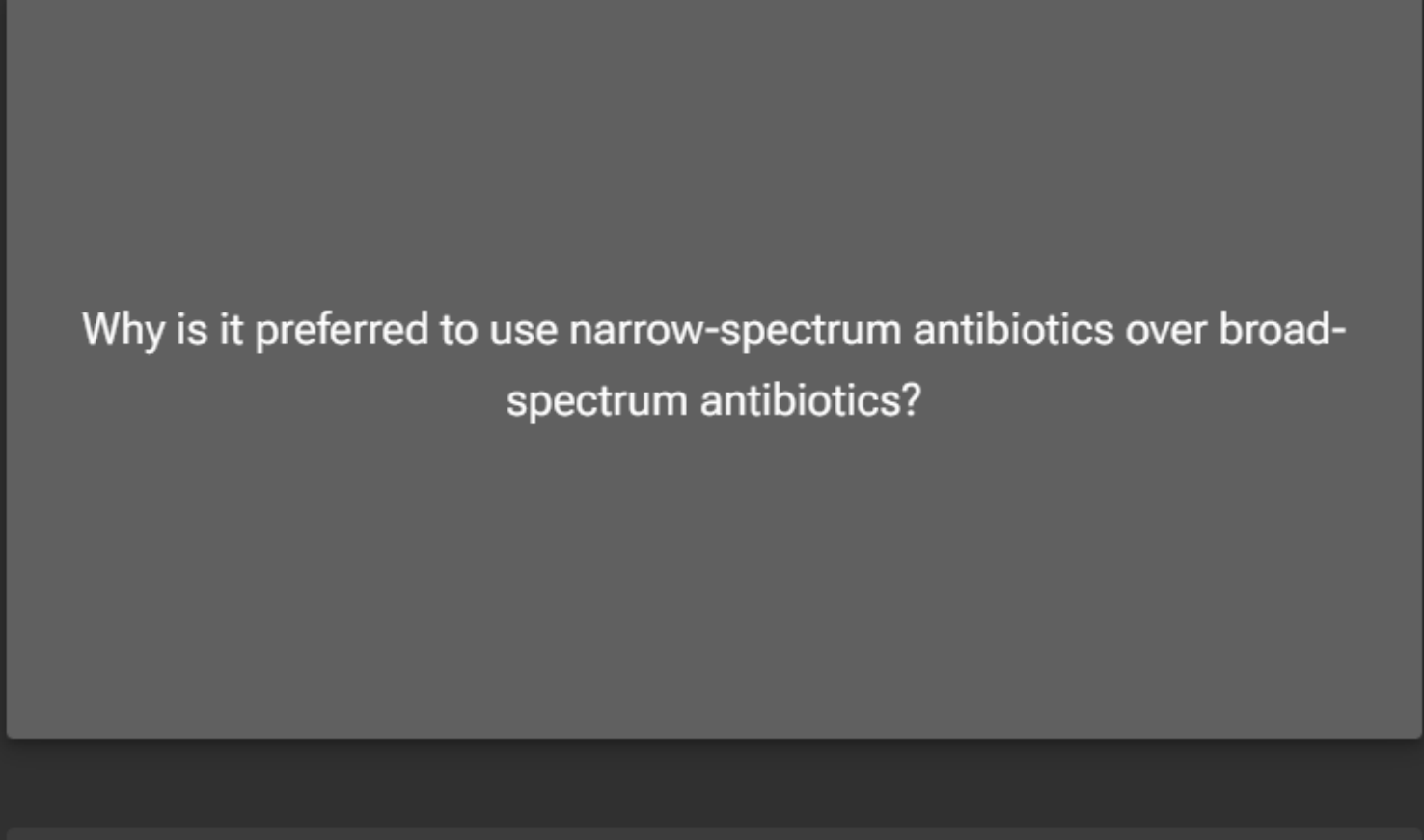
Suspect infection
Culture suspected sites
Gram stain
Identification
Susceptibilities
Begin empiric therapy for likely pathogens
Switch to definitive therapy for patient-specific pathog

Figure 1

Why is it preferred to use narrow-spectrum antibiotics over broad-spectrum antibiotics?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Returning to RP: He has a pneumonia with a multi-drug resistant organism, *Klebsiella*, which was resistant to his empiric therapy. By explaining to RP that once his samples were sent to the lab it takes about 48 hours to identify the organism and discover their susceptibilities. In the meantime, we start empiric therapy, which is our 'best guess' based on recent exposures (like his recent use of ciprofloxacin for UTI) and common pathogens. Often, we need to change coverage once we know exactly which organism we are treating. You would also explain that as bacteria are exposed to antibiotics, the more resistant strains might not be killed off like the more susceptible ones. This is called selection pressure and is a common bacterial m

Summary

- Antibiotic stewardship is crucial to optimize patient outcomes and minimize antibiotic resistance and adverse drug events.
- Bacteria can develop resistance through intrinsic resistance, selection pressure, acquired resistance, and enzyme inactivation.
- Antimicrobial Stewardship Programs (ASPs) in hospitals aim to optimize antibiotic use and minimize resistance development.
- Narrow-spectrum antibiotics are preferred over broad-spectrum ones to minimize microbiome disruption and reduce resistance risk.
- Empiric therapy involves broad-spectrum antibiotics based on suspected infections, while targeted therapy uses specific antibiotics once the pathogen is identified.

Review Questions

1. Which type of bacteria is capable of producing enzymes that breakdown penicillins and many cephalosporins, and are usually treated with carbapenem antibiotics?

A. ○ Beta-lactamses
B. ○ Carbapenem-resistant Enterobacteriaceae
C. ● Extended-
D. ○ Enterococcus

Hide Explanation

The correct answer is extended-spectrum beta-lactamases (C). Bacteria that produce extended-spectrum beta-lactamases are able to break down penicillins and most cephalosporins. The carbapenems are generally reserved to treat these bacteria. Many gram-negative bacteria produce beta-lactamses (A), but they are not necessarily extended spectrum. Carbapenem-resistent Enterobacteriaceae (B) are able to break down penicillins, cephalosporins, and carbapenems with their enzyme production. Enterococcus (D) is a gram-positive organism that does not produce beta-lactamses.

2. Why is narrow-spectrum antibiotic use preferred?

A. ● Minimizes microbiome disruption
B. ○ Kills more bacteria
C. ○ Increased resistance
D. ○ Broad-spectrum is preferred
E. ○ Cost effective

Hide Explanation

The correct answer is minimizes microbiome disruption (A). Narrow-spectrum antibiotics target specific bacteria, preserving the normal gut microbiome and reducing resistance risk. Broad-spectrum antibiotics (B) kill a wider range of bacteria, including beneficial ones. Increased resistance (C) is more likely with broad-spectrum use. Broad-spectrum use (D) is not preferred due to unnecessary resistance development. Cost (E) is not the primary reason for preference.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the major targets, common mechanisms of how bacteria become resistant to drugs, and explain strategies to combat antimicrobial resistance.~~

02 ☑ ~~Explain the rationale for the use of broad versus narrow spectrum and empiric versus targeted antibacterial therapy.~~

Objective Confidence: 2/2